2.1. Hanging rope

Let $T(y)$ be the tension as a function of height. Consider a small piece of the rope between y and $y + dy$ ($0 \le y \le L$). The forces on this piece are $T(y + dy)$ upward, $T(y)$ downward, and the weight $\rho g\, dy$ downward. Since the rope is at rest, we have $T(y + dy) = T(y) + \rho g\, dy$. Expanding this to first order in dy gives $T'(y) = \rho g$. The tension in the bottom of the rope is zero, so integrating from $y = 0$ up to a position y gives

$$T(y) = \rho g y. \tag{2.16}$$

As a double-check, at the top end we have $T(L) = \rho g L$, which is the weight of the entire rope, as it should be.

Alternatively, you can simply write down the answer, $T(y) = \rho g y$, by noting that the tension at a given point in the rope is what supports the weight of all the rope below it.

2.2. Block on a plane

Balancing the forces shown in Fig. 2.41, parallel and perpendicular to the plane, we see that $F = mg\sin\theta$ and $N = mg\cos\theta$. The horizontal components of these are $F\cos\theta = mg\sin\theta\cos\theta$ (to the right), and $N\sin\theta = mg\cos\theta\sin\theta$ (to the left).

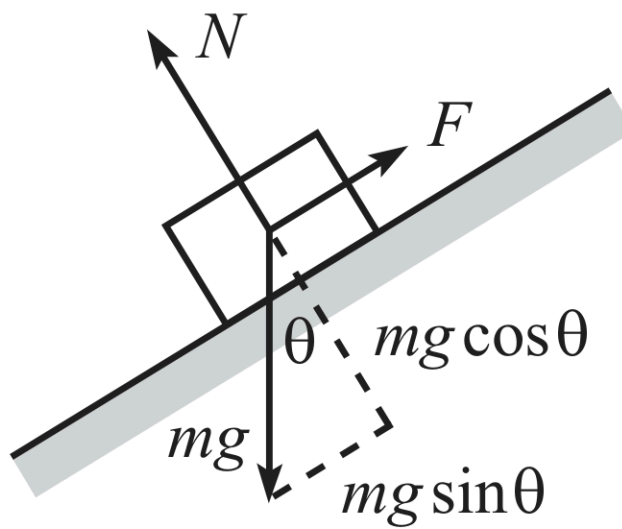


Fig. 2.41

These are equal, as they must be, because the net horizontal force on the block is zero. To maximize the value of $mg \sin\theta \cos\theta$, we can either take the derivative, or we can write it as $(mg/2) \sin 2\theta$, from which it is clear that the maximum occurs at $\theta = \pi/4$. The maximum value is $mg/2$.